Bishop 2024 - Deep Learning Ch 20 Diffusion Models Sec. 0,1,2

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Bishop, Christopher M., and Hugh Bishop. “Diffusion Models.” *Deep Learning*, by Christopher M. Bishop and Hugh Bishop, Springer International Publishing, 2024, pp. 581–607. *DOI.org (Crossref)*, https://doi.org/10.1007/978-3-031-45468-4_20.

Search Terms, Vocabulary, Classification

- Labels
  - Task: Generative distribution estimation
  - Model
    - Denoising Diffusion Probabilistic Models (DDPMs)
    - Hierarchical variational autoencoder
- Model
  - Probabilistic Graphical Model
    - Markov Chain
      - Latent Variable
      - Diffusion
        - Forward encoder
          - Fixed distribution converges to isotropic gaussian noise
        - Decoder
          - Deep neural network

Model selection

- Pros
  - Easy to train
  - Scalable, **parallelizable**
  - Avoids challenges, instabilities of incumbents, e.g. Generative Adversarial Networks (GANs), challenges such as:
    - Mode collapse
    - Vanishing gradients and divergence
    - Hyperparameter sensitivity
- Cons
  - Sample generation can be expensive

Probabilistic Graphical Model - Markov Chain - Latent Variable - Diffusion - Deep Network

$$= \mathcal{N}\left(\mathbf{z}_{t-1} | \mathbf{m}_t(\mathbf{x}, \mathbf{z}_t), \sigma_t^2 \mathbf{I}\right)$$

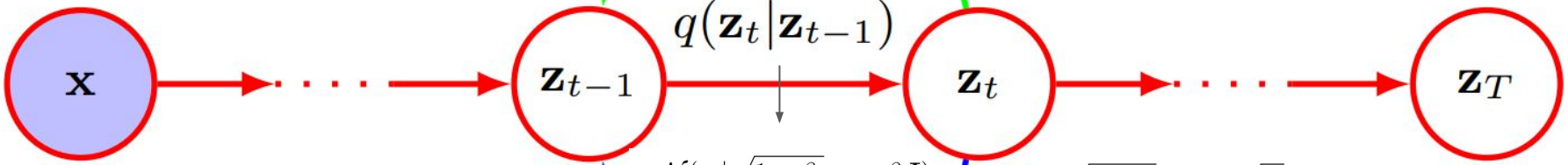
$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{(1-\alpha_{t-1})\sqrt{1-\beta_t}\mathbf{z}_t + \sqrt{\alpha_{t-1}}\beta_t \mathbf{x}}{1-\alpha_t}$$

$$\sigma_t^2 = \frac{\beta_t(1-\alpha_{t-1})}{1-\alpha_t}$$

$$\alpha_t = \prod_{\tau=1}^{t}(1-\beta_\tau).$$

$$q(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x})$$

$$q(\mathbf{z}_t | \mathbf{z}_{t-1})$$



$$= \mathcal{N}(\mathbf{z}_t | \sqrt{1-\beta_t}\mathbf{z}_{t-1}, \beta_t \mathbf{I})$$

$$= \sqrt{1-\beta_t}\mathbf{z}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t$$

$$= \sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t$$

$$= \mathcal{N}(\mathbf{z}_T | \mathbf{0}, \mathbf{I}).$$

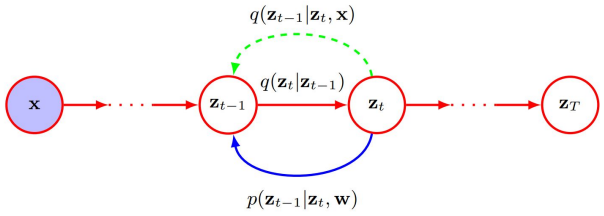
… reversible by deep neural network

$$p(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{w})$$

$$= \mathcal{N}(\mathbf{z}_{t-1} | \boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t), \beta_t \mathbf{I})$$

μ, g are (deep) neural networks

$$\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t)\right\}$$



The generator p(x,z|w) and noising q(z) = q(z|x) distributions then are:

$$p(\mathbf{x}, \mathbf{z}_1, \ldots, \mathbf{z}_T|\mathbf{w}) = p(\mathbf{z}_T) \left\{ \prod_{t=2}^{T} p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) \right\} p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})$$

$$q(\mathbf{z}_1, \ldots, \mathbf{z}_t|\mathbf{x}) = q(\mathbf{z}_1|\mathbf{x}) \prod_{\tau=2}^{t} q(\mathbf{z}_\tau|\mathbf{z}_{\tau-1}).$$

We would like to maximize the marginal of the data given the parameters (the likelihood) but we don't know how…

$$p(\mathbf{x}|\mathbf{w}) = \int \cdots \int p(\mathbf{x}, \mathbf{z}_1, \ldots, \mathbf{z}_T|\mathbf{w}) \, \mathrm{d}\mathbf{z}_1 \ldots \mathrm{d}\mathbf{z}_T$$

Instead, we'll derive an expression for a lower bound we know how to maximize…

By the product rule

$$p(\mathbf{x}, \mathbf{z}|\mathbf{w}) = p(\mathbf{z}|\mathbf{x}, \mathbf{w}) p(\mathbf{x}|\mathbf{w})$$

$$\mathcal{L}(\mathbf{w}) = \int q(\mathbf{z}) \ln \left\{ \frac{p(\mathbf{x}, \mathbf{z}|\mathbf{w})}{q(\mathbf{z})} \right\} \mathrm{d}\mathbf{z}$$

$$\mathrm{KL}\left(f(\mathbf{z}) \| g(\mathbf{z})\right) = -\int f(\mathbf{z}) \ln \left\{ \frac{g(\mathbf{z})}{f(\mathbf{z})} \right\} \mathrm{d}\mathbf{z}$$

$$\ln p(\mathbf{x}|\mathbf{w}) = \mathcal{L}(\mathbf{w}) + \mathrm{KL}\left(q(\mathbf{z}) \| p(\mathbf{z}|\mathbf{x}, \mathbf{w})\right)$$

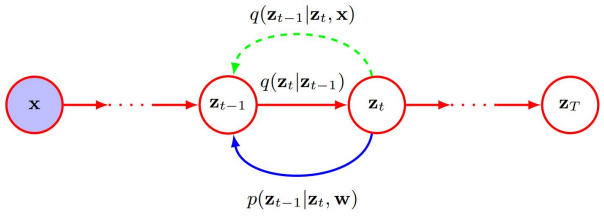
Combining product rule, L(w) and KL

Since KL > 0: $\ln p(\mathbf{x}|\mathbf{w}) \geqslant \mathcal{L}(\mathbf{w})$

Let $\mathbb{E}_q[\,\cdot\,] \equiv \int \cdots \int q(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q(\mathbf{z}_t|\mathbf{z}_{t-1}) [\,\cdot\,] \, \mathrm{d}\mathbf{z}_1 \ldots \mathrm{d}\mathbf{z}_T$

Expectation over q simplifies L(w)

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q \left[\ln \frac{p(\mathbf{z}_T) \left\{ \prod_{t=2}^{T} p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) \right\} p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})}{q(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} \right]$$



$$\mathbb{E}_q[\,\cdot\,] \equiv \int \cdots \int q(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q(\mathbf{z}_t|\mathbf{z}_{t-1}) [\,\cdot\,]\, \mathrm{d}\mathbf{z}_1 \ldots \mathrm{d}\mathbf{z}_T$$

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q \left[\ln \frac{p(\mathbf{z}_T) \left\{ \prod_{t=2}^{T} p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) \right\} p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})}{q(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} \right]$$

From last slide

Since ln(ab) = ln(a) + ln(b)

$$= \mathbb{E}_q \left[\ln p(\mathbf{z}_T) + \sum_{t=2}^{T} \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} - \ln q(\mathbf{z}_1|\mathbf{x}) + \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \right]$$

Log algebra

Constant wrt w

$$\mathbb{E}_q[\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})] \simeq \sum_{l=1}^{L} \ln p(\mathbf{x}|\mathbf{z}_1^{(l)}, \mathbf{w})$$

Inefficient monte carlo

Efficient monte carlo

Bayes

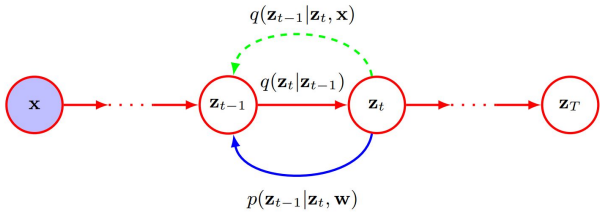
$$q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x}) = \frac{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) q(\mathbf{z}_t|\mathbf{x})}{q(\mathbf{z}_{t-1}|\mathbf{x})}$$

$$\ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} = \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} + \ln \frac{q(\mathbf{z}_{t-1}|\mathbf{x})}{q(\mathbf{z}_t|\mathbf{x})}$$

Log algebra

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q \left[\sum_{t=2}^{T} \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} + \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \right]$$

L now hinges on two terms, one estimable



Rewriting by definition of Eq, KL:

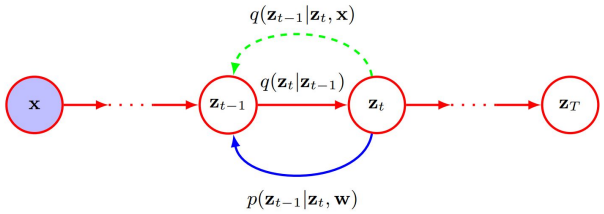
$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q \left[\boxed{\sum_{t=2}^{T} \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})}} + \boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})} \right]$$

$$\mathbb{E}_q[\,\cdot\,] \equiv \int \cdots \int \boxed{q(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q(\mathbf{z}_t|\mathbf{z}_{t-1})} [\,\cdot\,]\, \mathrm{d}\mathbf{z}_1 \ldots \mathrm{d}\mathbf{z}_T$$

$$\boxed{\mathrm{KL}\left(f(\mathbf{z}) \| g(\mathbf{z})\right) = -\int f(\mathbf{z}) \ln \left\{ \frac{g(\mathbf{z})}{f(\mathbf{z})} \right\} \mathrm{d}\mathbf{z}}$$

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int \boxed{q(\mathbf{z}_1|\mathbf{x})} \boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})}\, \mathrm{d}\mathbf{z}_1}_{\text{reconstruction term}}$$

$$\underbrace{- \sum_{t=2}^{T} \int \boxed{\mathrm{KL}\left(\boxed{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} \| p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})\right)} \boxed{q(\mathbf{z}_t|\mathbf{x})}\, \mathrm{d}\mathbf{z}_t}_{\text{consistency terms}}$$



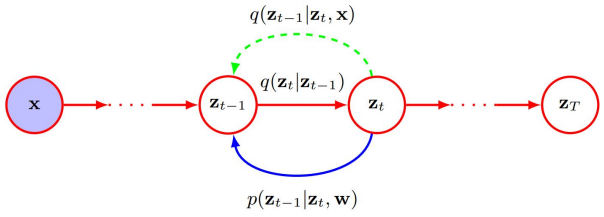
$$\mathbb{E}_q\left[\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})\right] \simeq \sum_{l=1}^{L} \ln p(\mathbf{x}|\mathbf{z}_1^{(l)}, \mathbf{w})$$

Efficient monte carlo

The reconstruction term, depending on a single zt can be estimated efficiently using monte carlo as for 4th term in Eq. $q(z_t|x)$ (above) is estimable as well. KL requires additional manipulation due to dependence on multiple z.

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})\, \mathrm{d}\mathbf{z}_1}_{\text{reconstruction term}} - \underbrace{\sum_{t=2}^{T} \int \mathrm{KL}\langle q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \| p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})\rangle q(\mathbf{z}_t|\mathbf{x})\, \mathrm{d}\mathbf{z}_t}_{\text{consistency terms}}$$

$$q(\mathbf{z}_t|\mathbf{x}) = \mathcal{N}(\mathbf{z}_t|\sqrt{\alpha_t}\mathbf{x}, (1-\alpha_t)\mathbf{I})$$



$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})} \, \mathrm{d}\mathbf{z}_1}_{\text{reconstruction term}}$$

$$\underbrace{- \sum_{t=2}^{T} \int \mathrm{KL}(\boxed{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} \| \boxed{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}) q(\mathbf{z}_t|\mathbf{x}) \, \mathrm{d}\mathbf{z}_t}_{\text{consistency terms}}$$

Making KL estimable:

$$\mathrm{KL}(\boxed{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} \| \boxed{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})})$$

$$\mathrm{KL}\left(f(\mathbf{z}) \| \boxed{g(\mathbf{z})}\right) = -\int \boxed{f(\mathbf{z})} \ln \left\{ \frac{\boxed{g(\mathbf{z})}}{\boxed{f(\mathbf{z})}} \right\} \mathrm{d}\mathbf{z} = \mathbb{E}_f[\ln f(\mathbf{z})] - \mathbb{E}_f[\ln g(\mathbf{z})]$$

Definition of expectation

$$\ln f(\mathbf{z}) = -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_1| - \frac{1}{2} (\mathbf{z} - \boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}_1^{-1} (\mathbf{z} - \boldsymbol{\mu}_1)$$

Log normal density

$$\mathbb{E}_f[\ln f(\mathbf{z})] = -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_1| - \frac{d}{2}$$

Expectation of f over f

$$\mathbb{E}_f[\ln g(\mathbf{z})] = -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_2| - \frac{1}{2} \mathbb{E}_f[(\mathbf{z} - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1} (\mathbf{z} - \boldsymbol{\mu}_2)]$$

Expectation of g over f

$$\mathbb{E}_f[(\mathbf{z} - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1} (\mathbf{z} - \boldsymbol{\mu}_2)] = \mathrm{Tr}(\boldsymbol{\Sigma}_2^{-1} \boldsymbol{\Sigma}_1) + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

$$D_{KL}(f \parallel g) = \left[-\frac{1}{2} \ln |\boldsymbol{\Sigma}_1| - \frac{d}{2} \right] - \left[-\frac{1}{2} \ln |\boldsymbol{\Sigma}_2| - \frac{1}{2} \mathrm{Tr}(\boldsymbol{\Sigma}_2^{-1} \boldsymbol{\Sigma}_1) - \frac{1}{2} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \right]$$

Combining

$$D_{KL}(f \parallel g) = \frac{1}{2} \left[\ln \frac{|\boldsymbol{\Sigma}_2|}{|\boldsymbol{\Sigma}_1|} - d + \mathrm{Tr}(\boldsymbol{\Sigma}_2^{-1} \boldsymbol{\Sigma}_1) + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \right]$$

Simplifying

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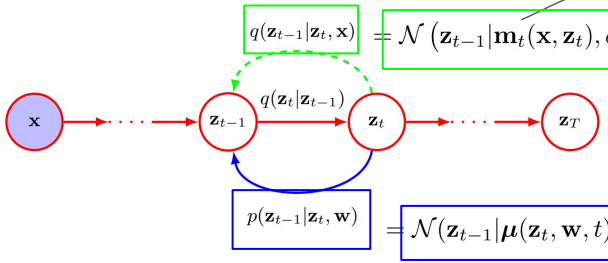

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \mathcal{N}\left(\mathbf{z}_{t-1}|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t), \sigma_t^2 \mathbf{I}\right)$$

$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{(1-\alpha_{t-1})\sqrt{1-\beta_t}\mathbf{z}_t + \sqrt{\alpha_{t-1}}\beta_t \mathbf{x}}{1-\alpha_t}$$

$$\sigma_t^2 = \frac{\beta_t(1-\alpha_{t-1})}{1-\alpha_t}$$

$$\alpha_t = \prod_{\tau=1}^{t}(1-\beta_\tau).$$

$$\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t)\right\}$$

$$p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) = \mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t), \beta_t \mathbf{I})$$

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})\, \mathrm{d}\mathbf{z}_1}_{\text{reconstruction term}} - \underbrace{\sum_{t=2}^{T} \int \mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \| p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}))\, q(\mathbf{z}_t|\mathbf{x})\, \mathrm{d}\mathbf{z}_t}_{\text{consistency terms}}$$

Making KL efficiently estimable:

$$D_{KL}(f \| g) = \frac{1}{2}\left[\ln \frac{|\boldsymbol{\Sigma}_2|}{|\boldsymbol{\Sigma}_1|} - d + \mathrm{Tr}(\boldsymbol{\Sigma}_2^{-1}\boldsymbol{\Sigma}_1) + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\right]$$

Simplifying

$$= \frac{1}{2\beta_t}\|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) - \boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t)\|^2 + \text{const}$$

Simplifying and plugging in m, μ

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Now L(w) is fully estimable. But we can do better:

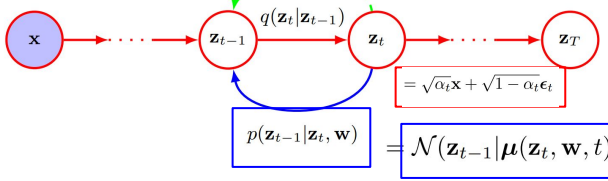
$$q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \mathcal{N}\left(\mathbf{z}_{t-1}|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t), \sigma_t^2 \mathbf{I}\right)$$

$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{(1-\alpha_{t-1})\sqrt{1-\beta_t}\mathbf{z}_t + \sqrt{\alpha_{t-1}}\beta_t \mathbf{x}}{1-\alpha_t}$$

$$\sigma_t^2 = \frac{\beta_t(1-\alpha_{t-1})}{1-\alpha_t}$$

$$\alpha_t = \prod_{\tau=1}^{t}(1-\beta_\tau).$$

$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\boldsymbol{\epsilon}_t\right\}.$$



$$\boxed{\mathbf{x} = \frac{1}{\sqrt{\alpha_t}}\mathbf{z}_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t.}$$

$$= \boxed{\mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t), \beta_t \mathbf{I})}$$

$$\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t)\right\}$$

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})} \, d\mathbf{z}_1}_{\text{reconstruction term}}$$

$$\underbrace{-\sum_{t=2}^{T} \int \mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \boxed{\| p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}) q(\mathbf{z}_t|\mathbf{x}) \, d\mathbf{z}_t}_{\text{consistency terms}}$$

Reconstruction:

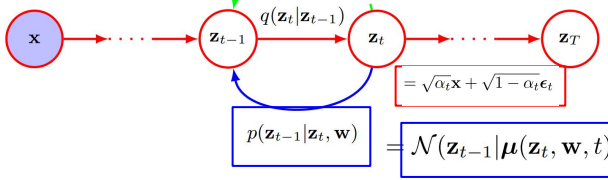
Expand and simplify log normal density:

$$\boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) =} -\frac{1}{2\beta_1}\|\mathbf{x} - \boldsymbol{\mu}(\mathbf{z}_1, \mathbf{w}, 1)\|^2 + \text{const.}$$

And substituting mu, x, alpha:

$$\boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})} = -\frac{1}{2(1-\beta_t)}\|\mathbf{g}(\mathbf{z}_1, \mathbf{w}, 1) - \boldsymbol{\epsilon}_1\|^2 + \text{const.}$$

Now L(w) is fully estimable. But we can do better:



$$= \mathcal{N}\left(\mathbf{z}_{t-1}|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t), \sigma_t^2\mathbf{I}\right)$$

$$= \mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t), \beta_t\mathbf{I})$$

$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{(1-\alpha_{t-1})\sqrt{1-\beta_t}\mathbf{z}_t + \sqrt{\alpha_{t-1}}\beta_t\mathbf{x}}{1-\alpha_t}$$

$$\sigma_t^2 = \frac{\beta_t(1-\alpha_{t-1})}{1-\alpha_t}$$

$$\alpha_t = \prod_{\tau=1}^{t}(1-\beta_\tau).$$

$$\mathbf{x} = \frac{1}{\sqrt{\alpha_t}}\mathbf{z}_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t.$$

$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\boldsymbol{\epsilon}_t\right\}.$$

$$\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t)\right\}$$

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})}\, d\mathbf{z}_1}_{\text{reconstruction term}} - \underbrace{\sum_{t=2}^{T} \int \mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \boxed{\|p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}) q(\mathbf{z}_t|\mathbf{x})\, d\mathbf{z}_t}_{\text{consistency terms}}$$

Reconstruction:

Expand and simplify log normal density:

$$\boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) =} -\frac{1}{2\beta_1}\|\mathbf{x} - \boldsymbol{\mu}(\mathbf{z}_1, \mathbf{w}, 1)\|^2 + \text{const.}$$

And substituting mu, x, alpha:

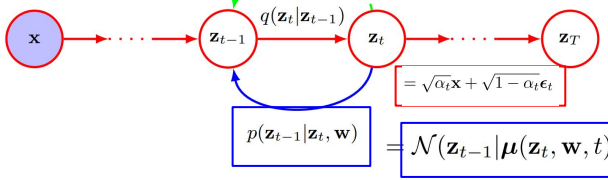
$$\boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})} = -\frac{1}{2(1-\beta_t)}\|\mathbf{g}(\mathbf{z}_1, \mathbf{w}, 1) - \boldsymbol{\epsilon}_1\|^2 + \text{const.}$$

$$\mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})\|p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}))$$
$$= \frac{1}{2\beta_t}\|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) - \boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t)\|^2 + \text{const}$$

Plugging in m, x and μ to latest KL consistency expression:

$$\mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})\|p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}))$$
$$= \frac{\beta_t}{2(1-\alpha_t)(1-\beta_t)}\|\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}_t\|^2 + \text{const}$$
$$= \frac{\beta_t}{2(1-\alpha_t)(1-\beta_t)}\left\|\mathbf{g}(\sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}_t\right\|^2 + \text{const}$$

Now L(w) is fully estimable. But we can do better:



$$= \mathcal{N}\left(\mathbf{z}_{t-1}|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t), \sigma_t^2\mathbf{I}\right)$$

$$= \mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t), \beta_t\mathbf{I})$$

$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{(1-\alpha_{t-1})\sqrt{1-\beta_t}\mathbf{z}_t + \sqrt{\alpha_{t-1}}\beta_t\mathbf{x}}{1-\alpha_t}$$

$$\sigma_t^2 = \frac{\beta_t(1-\alpha_{t-1})}{1-\alpha_t}$$

$$\alpha_t = \prod_{\tau=1}^{t}(1-\beta_\tau).$$

$$\mathbf{x} = \frac{1}{\sqrt{\alpha_t}}\mathbf{z}_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t.$$

$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\boldsymbol{\epsilon}_t\right\}.$$

$$\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) = \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t)\right\}$$

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})}\, d\mathbf{z}_1}_{\text{reconstruction term}} - \underbrace{\sum_{t=2}^{T}\int \mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})\|\boxed{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})})q(\mathbf{z}_t|\mathbf{x})\, d\mathbf{z}_t}_{\text{consistency terms}}$$

$$\mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})\|p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}))$$
$$= \frac{1}{2\beta_t}\|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) - \boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t)\|^2 + \text{const}$$

$$q(\mathbf{z}_t|\mathbf{x}) = \mathcal{N}(\mathbf{z}_t|\sqrt{\alpha_t}\mathbf{x}, (1-\alpha_t)\mathbf{I})$$

Reconstruction:

Expand and simplify log normal density:

$$\boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) =} -\frac{1}{2\beta_1}\|\mathbf{x} - \boldsymbol{\mu}(\mathbf{z}_1, \mathbf{w}, 1)\|^2 + \text{const.}$$

And substituting mu, x, alpha:

$$\boxed{\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})} = -\frac{1}{2(1-\beta_t)}\|\mathbf{g}(\mathbf{z}_1, \mathbf{w}, 1) - \boldsymbol{\epsilon}_1\|^2 + \text{const.}$$

Plugging in m, x and μ to latest KL consistency expression:

$$\mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})\|p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}))$$
$$= \frac{\beta_t}{2(1-\alpha_t)(1-\beta_t)}\|\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}_t\|^2 + \text{const}$$
$$= \frac{\beta_t}{2(1-\alpha_t)(1-\beta_t)}\left\|\mathbf{g}(\sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}_t\right\|^2 + \text{const}$$

Empirically Ho, Jain and Abbeel 2020 found that estimates improved without the leading coefficient on the norm, omitting it yields:

$$\mathcal{L}(\mathbf{w}) = -\sum_{t=1}^{T}\left\|\mathbf{g}(\sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}_t\right\|^2$$

Training

- Finding the deep neural network weights w which maximize the ELBO, thereby maximizing the likelihood of the data given them
- Steps
  - For each latent, calculate drift/shift coefficients from the data
  - Sample
    - Data
    - Latent
    - Noise (isotropic)
  - Calculate
    - Latent
    - Loss
    - Update to w

Algorithm 20.1: Training a denoising diffusion probabilistic model

Input: Training data $\mathcal{D} = \{\mathbf{x}_n\}$
Noise schedule $\{\beta_1, \ldots, \beta_T\}$
Output: Network parameters $\mathbf{w}$

---

for $t \in \{1, \ldots, T\}$ **do**
  $\alpha_t \leftarrow \prod_{\tau=1}^{t}(1-\beta_\tau)$ `// Calculate alphas from betas`
end for
repeat
  $\mathbf{x} \sim \mathcal{D}$ `// Sample a data point`
  $t \sim \{1, \ldots, T\}$ `// Sample a point along the Markov chain`
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}|\mathbf{0}, \mathbf{I})$ `// Sample a noise vector`
  $\mathbf{z}_t \leftarrow \sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}$ `// Evaluate noisy latent variable`
  $\mathcal{L}(\mathbf{w}) \leftarrow \|\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}\|^2$ `// Compute loss term`
  Take optimizer step
until converged
return $\mathbf{w}$

Sampling

- Generating new data from the trained model
- Steps
  - Sample latent noise
  - Iteratively
    - Calculate data-drift-shift coefficient
    - Evaluate neural net transformation
    - Sample and add noise
  - Denoise

Algorithm 20.2: Sampling from a denoising diffusion probabilistic model

Input: Trained denoising network $\mathbf{g}(\mathbf{z}, \mathbf{w}, t)$
Noise schedule $\{\beta_1, \ldots, \beta_T\}$
Output: Sample vector $\mathbf{x}$ in data space

$\mathbf{z}_T \sim \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I})$ `// Sample from final latent space`
for $t \in T, \ldots, 2$ **do**
  $\alpha_t \leftarrow \prod_{\tau=1}^{t}(1-\beta_\tau)$ `// Calculate alpha`
  `// Evaluate network output`
  $\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) \leftarrow \frac{1}{\sqrt{1-\beta_t}}\left\{\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t)\right\}$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}|\mathbf{0}, \mathbf{I})$ `// Sample a noise vector`
  $\mathbf{z}_{t-1} \leftarrow \boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) + \sqrt{\beta_t}\boldsymbol{\epsilon}$ `// Add scaled noise`
end for
$\mathbf{x} = \frac{1}{\sqrt{1-\beta_1}}\left\{\mathbf{z}_1 - \frac{\beta_1}{\sqrt{1-\alpha_1}}\mathbf{g}(\mathbf{z}_1, \mathbf{w}, t)\right\}$ `// Final denoising step`
return $\mathbf{x}$